

This is the fifth in a series of 10-year updates in child and adolescent psychiatry. Topics are selected in consultation with the AACAP Committee on Recertification, both for the importance of new research and its clinical or developmental significance. The authors have been asked to place an asterisk before the five or six most seminal references.

J. McD.

Childhood and Adolescent Depression: A Review of the Past 10 Years. Part I

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ABSTRACT

Objective: To qualitatively review the literature of the past decade covering the epidemiology, clinical characteristics, natural course, biology, and other correlates of early-onset major depressive disorder (MDD) and dysthymic disorder (DD). **Method:** A computerized search for articles published during the past 10 years was made and selected studies are presented. **Results:** Early-onset MDD and DD are frequent, recurrent, and familial disorders that tend to continue into adulthood, and they are frequently accompanied by other psychiatric disorders. These disorders are usually associated with poor psychosocial and academic outcome and increased risk for substance abuse, bipolar disorder, and suicide. In addition, DD increases the risk for MDD. There is a secular increase in the prevalence of MDD, and it appears that MDD is occurring at an earlier age in successive cohorts. Several genetic, familial, demographic, psychosocial, cognitive, and biological correlates of onset and course of early-onset depression have been identified. Few studies, however, have examined the combined effects of these correlates. **Conclusions:** Considerable advances have been made in our knowledge of early-onset depression. Nevertheless, further research is needed in understanding the pathogenesis of childhood mood disorders. Toward this end, studies aimed at elucidating mechanisms and interrelationships among the different domains of risk factors are needed. *J. Am. Acad. Child Adolesc. Psychiatry*, 1996, 35(11):1427–1439. **Key Words:** children, adolescents, major depressive disorder, dysthymic disorder, progress, correlates.

This article will review selected articles published during the past decade that focused on epidemiology, clinical characteristics, natural history, and several correlates of early-onset depression. A second part, to be published in a subsequent issue of this *Journal*, will review the

past decade of literature on assessment, treatment, and prevention of early-onset depression.

EPIDEMIOLOGY

Population studies of children and adolescents have reported prevalence rates of depression in children ranging between 0.4% and 2.5% in children and between 0.4% and 8.3% in adolescents (Anderson and McGee, 1994; Fleming and Offord, 1990; Kashani et al., 1987a,b; Lewinsohn et al., 1986, 1993a, 1994). The lifetime prevalence rate of major depressive disorder (MDD) in adolescents has been estimated to range from 15% to 20%, which is comparable with the lifetime rate of MDD found in adult populations, suggesting that depression in adults often begins in adolescence (Kessler et al., 1994b; Lewinsohn et al., 1986, 1993a,c).

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The few epidemiological studies on dysthymic disorder (DD) have reported a point prevalence rate from 0.6% to 1.7% in children and 1.6% to 8.0% in adolescents (Kashani et al., 1987a,b; Lewinsohn et al., 1993a, 1994).

In children, MDD occurs at approximately the same rate in girls and in boys, whereas in adolescents, the female-to-male ratio is approximately 2:1, paralleling the ratio reported in adult MDD (Fleming and Offord, 1990; Kessler et al., 1994a; Lewinsohn et al., 1994). While the nature of this sex difference is as yet unclear, it has been attributed to genetics, increased prevalence of anxiety disorders in females, biological changes associated with puberty, cognitive predisposition, and socio-cultural factors (Breslau et al., 1995; Reinherz et al., 1989; Rutter, 1991).

Secular Increase. Investigations of clinic-referred and population samples of adults and children with mood disorders have reported that individuals born in the latter part of the 20th century are at greater risk for developing mood disorders and that these disorders are manifesting at a younger age (e.g., Gershon et al., 1987; Joyce et al., 1990; Kessler et al., 1994a; Klein et al., 1995; Kovacs and Gatsonis, 1994; Lavori et al., 1987; Lewinsohn et al., 1993c; Ryan et al., 1992b; Wickramaratne et al., 1989). This trend, however, appears to apply for mild to moderate depressions and not for more severe melancholic depressions (Hagnell et al., 1982) and has not been found in DD (Klein et al., 1995). The reason for the secular increase is not yet clear, but since the genetic makeup of the population has not changed substantially, the increase is most likely due to either environmental factors or the interrelation of environmental and genetic factors (Gershon et al., 1987; Klerman and Weissman, 1988).

CLINICAL CHARACTERISTICS

Major Depressive Disorder. The clinical picture of early-onset MDD parallels the phenomenology of adult MDD (*DSM-IV*) (American Psychiatric Association, 1994) (e.g., Roberts et al., 1995; Ryan et al., 1987). However, there are some developmental differences. Symptoms of endogeneity/melancholia, psychosis, suicide attempts, lethality of suicide attempt, and impairment of functioning increase with age. In contrast, symptoms of separation anxiety, phobias, somatic complaints, and behavioral problems seem to occur more frequently in children (e.g., Carlson and Kashani, 1988;

Kolvin et al., 1991; Mitchell et al., 1988; Ryan et al., 1987). Psychotic depression in children appears to be manifested by auditory hallucinations instead of delusions as seen in adolescents and adults, and it has been attributed to the lack of cognitive maturation in children (e.g., Ryan et al., 1987). Seasonal affective disorder, atypical depression, and premenstrual dysphoric disorder tend to emerge during adolescence and may require different treatment strategies, but these disorders have not been well studied in youth (Lucas, 1991; Stewart et al., 1993; Swedo et al., 1995).

Dysthymic Disorder. Except for the requirement of 1 year duration instead of 2 years and that children may have only irritable mood instead of depression, the *DSM* criterion for early-onset and adult DD are identical (*DSM-IV*) (American Psychiatric Association, 1994). Other symptoms such as feelings of being unloved, anger, self-deprecation, somatic complaints, anxiety, and disobedience have been reported in early-onset DD (Kovacs et al., 1994a). Furthermore, in contrast to the *DSM* standard, children with DD appear to have few melancholic symptoms compared with those with MDD (Kovacs et al., 1994a). Approximately 70% of individuals with early-onset DD eventually will develop an episode of MDD, resulting in the presence of both diagnoses, the so-called "double depression" (e.g., Ferro et al., 1994; Kovacs et al., 1994a; Lewinsohn et al., 1991).

Comorbidity

Major Depressive Disorder. Clinical (e.g., Biederman et al., 1995a; Kovacs et al. 1984a,b; Puig-Antich and Rabinovich, 1986; Ryan et al., 1987) as well as epidemiological investigations (Anderson and McGee, 1994; Angold and Costello, 1993; Bird et al., 1988; Kashani et al., 1987a,b; Rohde et al., 1991) have shown that 40% to 70% of depressed children and adolescents have comorbid psychiatric disorders, and at least 20% to 50% have two or more comorbid diagnoses. The most frequent comorbid diagnoses are DD and anxiety disorders (both at 30% to 80%), disruptive disorders (10% to 80%), and substance abuse (20% to 30%). Except for substance abuse, MDD is more likely to occur after the onset of other psychiatric disorders (e.g., Biederman et al., 1995a; Kovacs et al., 1989; Reinherz et al., 1993). However, conduct problems may develop as a complication of the depression and persist after the depression remits (Kovacs et al., 1988).

A few studies have reported that more than 60% of depressed adolescents have comorbid personality disorders, with borderline personality disorder accounting for 30% of all comorbid personality disorders (Marton et al., 1989). After the depression has remitted, however, personality disorder symptoms are no longer evident (Marton et al., 1987), highlighting the importance of giving only provisional personality disorder diagnoses during acute depressive episodes.

Dysthymic Disorder. Approximately 70% of the early-onset DD patients have a superimposed MDD and 50% have other preexisting psychiatric disorders, including anxiety disorders (40%), conduct disorder (30%), ADHD (24%), and enuresis or encopresis (15%), with 15% having two or more comorbid disorders (Kovacs et al., 1994a).

Comorbidity: Clinical and Functional Implications

In general, comorbid diagnoses appear to influence the risk for recurrent depression, duration of the depressive episode, suicide attempts or behaviors, functional outcome, response to treatment, and utilization of mental health services (Brent et al., 1988, 1993a,b, 1994; Clarke et al., 1992; Kovacs et al., 1993; Lewinsohn et al., 1993b, 1994, 1995b; Marton et al., 1989; Rohde et al., 1991; Sanford et al., 1995). In particular, youths with "double depressions" (MDD and DD) have been found to have more severe and longer depressive episodes, a higher rate of comorbid disorders, more suicidality, and worse social impairment than youths with MDD or DD alone (Ferro et al., 1994; Kovacs et al., 1994a; Lewinsohn et al., 1991). The comorbidity of depression and anxiety may also have clinical and functional implications as evidenced by studies showing an increased severity and duration of depressive symptoms, increased risk for substance abuse, increased suicidality, poor response to psychotherapy, and more psychosocial problems (e.g., Brent et al., 1988, 1993a; Clarke et al., 1992; Kendall et al., 1992; Kovacs et al., 1989). Depressed patients with comorbid disruptive disorders tend to have worse short-term outcome, fewer melancholic symptoms, fewer recurrences of depression, a lower familial aggregation of mood disorders, a higher incidence of adult criminality, more suicide attempts, higher levels of family criticism, and a higher response to placebo than MDD patients without disruptive disorders (Asarnow et al., 1994; Biederman et al., 1991; Harrington et al., 1990, 1991;

Hughes et al., 1990; Kutcher et al., 1989; Puig-Antich et al., 1989b). These findings suggest that depressed children with disruptive disorders may comprise a distinct etiological subgroup.

NATURAL COURSE

Major Depressive Disorder. Clinical and epidemiological studies in children and adolescents have reported that the mean length of an episode of MDD is approximately 7 to 9 months (Kovacs et al., 1984b; Lewinsohn et al., 1994; McCauley et al., 1993; Rao et al., 1995; Strober et al., 1993; Warner et al., 1992). Approximately 90% of the major depressive episodes have remitted by 1.5 to 2 years after the onset, with 6% to 10% becoming protracted (Kovacs et al., 1984a,b; McCauley et al., 1993; Sanford et al., 1995; Strober et al., 1993). Longitudinal studies of clinical (Kovacs et al., 1984a,b; McCauley et al., 1993; McGee and Williams, 1988; Rao et al., 1995; Sanford et al., 1995; Strober and Carlson, 1982; Strober et al., 1993) as well as epidemiological samples (Fleming et al., 1993; Hammen et al., 1990a; Lewinsohn et al., 1994; Warner et al., 1992) have consistently found that MDD is a recurrent condition with a cumulative probability of recurrence of 40% by 2 years and 70% by 5 years. Investigations of adolescents going into adulthood (Rao et al., 1995) and adult patients using catch-up longitudinal designs where initial diagnoses have been retrospectively reconstructed from clinical data summaries (Garber et al., 1988; Harrington et al., 1990; King and Pittman, 1970) have also shown that depression persists into adulthood, with recurrence rates estimated to be 60% to 70%. These results are identical with the 70% rate of recurrence reported in a 5-year prospective follow-up of adults with unipolar depression (Coryell et al., 1989). Very few studies have investigated the psychosocial and biological factors that may contribute to the recurrence of childhood MDD (e.g., Asarnow and Ben-Meir, 1988; Warner et al., 1992). For example, it has been reported that depressed children who live in conflictive family environments have higher recurrence rates than those who live in families with less conflict (Asarnow and Ben-Mier, 1988; Asarnow et al., 1994).

Risk of Developing Bipolar I and II Disorders. Follow-up studies have found that 20% to 40% of adolescents with MDD develop bipolar I disorder (periods of MDD and mania) within a period of 5 years after the

onset of depression (Garber et al., 1988; Geller et al., 1994; Kovacs and Gatsonis, 1989; Rao et al., 1995; Strober and Carlson, 1982; Strober et al., 1993). Clinical characteristics associated with an increased risk of developing bipolar I disorder in adolescents and adults with MDD include early-onset depression, depression accompanied by psychomotor retardation or psychotic features, family history of bipolar disorder or heavy loading for mood disorders, and pharmacologically induced hypomania (e.g., Akiskal et al., 1995; Geller et al., 1994; Strober and Carlson, 1982). In young depressive adults, the conversion to bipolar II disorder (periods of MDD and hypomania) has been associated with early-onset depression, atypical depression, seasonal affective disorder, protracted depressive episodes, mood lability, comorbid substance abuse, and high rates of psychosocial problems (e.g., Akiskal et al., 1995; Brent et al., 1988, 1993b; Lewinsohn et al., 1995a). It is important to recognize the existence of bipolar II disorder in adolescents because its clinical presentation may be easily misdiagnosed as a disruptive disorder or a personality disorder, particularly borderline personality disorder.

Dysthymic Disorder. Early-onset DD has a protracted course, with a mean episode length of about 4 years, and is associated with an increased risk for subsequent MDD (70%), bipolar disorder (13%), and substance abuse (15%) (Keller et al., 1988; Kovacs et al., 1984a,b, 1994a; Lewinsohn et al., 1991). Dysthymic children usually have their first episode of MDD 2 to 3 years after the onset of DD, suggesting that DD is one of the "gateways" to the development of recurrent mood disorders and indicating the need to develop preventive interventions targeted at this population (Kovacs et al., 1994a).

SEQUELAE

During the episode of depression, children and adolescents with clinical depression frequently experience impairment in school performance and relationships with others (e.g., Asarnow et al., 1987, 1990; Asarnow and Ben-Meir, 1988; Hammen, 1990; Kashani et al., 1988; Puig-Antich et al., 1985a,b, 1993; Rao et al., 1995; Strober et al., 1993; Williamson et al., 1995c). However, these psychosocial disturbances do not appear to be specific for depression in children and adolescents since they have been observed in youths with other psychopathology (e.g., Puig-Antich et al. 1985a). Also,

longitudinal studies are needed to document whether these psychosocial disturbances are sequelae or precursors of the disorders. Furthermore, independent of the depression, other factors, such as comorbid psychiatric disorders, poor family functioning, low socioeconomic status, and exposure to stressful life events impact the psychosocial functioning of the depressed patients, emphasizing the importance of managing these problems in addition to treating the depression (e.g., Asarnow et al., 1994; Marton et al., 1989; McCauley and Myers, 1992; Warner et al., 1995). Depression in children and adolescents is also associated with an increased risk of suicidal behaviors, homicidal ideation, tobacco use, and abuse of alcohol and other substances during later adolescence (Deykin et al., 1992; Kandel and Davies, 1986) and adulthood (Rao et al., 1995). In general, MDD precedes the onset of alcohol or substance abuse by an average of 4.5 years, providing a window of opportunity for the prevention of substance abuse in depressed adolescents.

Prospective studies have also found that after recovery, children and adolescents may continue to show subclinical symptoms of depression, negative attributions, impairment in interpersonal relationships, increased smoking, impairment in global functioning, early pregnancy, and increased physical problems (Kandel and Davies, 1986; Kovacs et al., 1994b; Nolen-Hoeksema et al., 1992; Puig-Antich et al., 1985a,b, 1993; Rao et al., 1995; Rohde et al., 1994; Strober et al., 1993). It is interesting that adolescents with two or more depressive episodes appear to have poorer functioning, while adolescents with nonrecurrent MDD may have good psychosocial outcomes similar to normal controls (Rao et al., 1995; Warner et al., 1995).

Suicide and Suicide Attempts. Paralleling the secular increase in MDD, the adolescent suicide rate has quadrupled since 1950 (2.5 to 11.2×100.000) and currently represents 12% of the total mortality in this age group (e.g., Brent et al., 1988; Lewinsohn et al., 1993b). Similarly, adolescent suicide attempts have also increased in recent years and have been found to have 1-year and lifetime prevalence rates of 1.7% to 5.9% and 3.0% to 7.1%, respectively (Centers for Disease Control, 1994; Fergusson and Lynskey, 1995; Lewinsohn et al., 1993b). These findings underscore the need for an accurate evaluation of suicidality and the implementation of preventive interventions. Beyond depression, predisposing factors for suicidality include

anxiety, disruptive, bipolar, substance abuse, and personality disorders. In addition, family history of mood disorders, family history of suicidal behavior, exposure to family violence, impulsivity, and availability of methods (e.g., firearms) have been associated with an increased risk for suicide (e.g., Brent et al., 1987, 1993a,b; Fergusson and Lynskey, 1995; Kovacs et al., 1993; Lewinsohn et al., 1994, 1995b; Pfeffer et al., 1993; Shaffer et al., 1996).

FACTORS ASSOCIATED WITH THE ONSET, DURATION, AND RECURRENCE OF EARLY-ONSET DEPRESSION

Twin and adoption studies in adult populations have provided evidence that genetic factors account for at least 50% of the variance in the transmission of mood disorders. Genetic studies have also suggested the importance of the impact of environmental factors, particularly nonshared intra- and extrafamilial environmental experiences including differences in the ways in which individual parents treat each of their children (Kendler, 1995; Plomin, 1994). Individuals at high genetic risk appear to be more sensitive to the effects of adverse environment than individuals at low genetic risk (Kendler, 1995). Furthermore, it has been suggested that environmental effects may be, at least in part, under genetic influence (Plomin, 1994).

Numerous factors have been associated with the onset, duration, and recurrence of early-onset depression, including the following: demographic factors (e.g., age, gender, socioeconomic status); psychopathology (e.g., preexisting diagnosis, subsyndromal depressive symptoms, negative cognitive style); familial factors (e.g., parental psychopathology, early-onset mood disorders, high familial loading for mood disorders); and psychosocial factors (e.g., poor support, stressful life events, poor maternal functioning). These factors appear to influence differentially the onset and natural course of the disorder (e.g., Kovacs et al., 1984b; Reinherz et al., 1993; Sanford et al., 1995). Several of these factors are briefly reviewed below.

Family Aggregation Studies

Major Depressive Disorder: "Top-Down" Studies of Children of Depressed Parents. Overall, children of depressed parents are three times more likely to have a lifetime episode of MDD. The lifetime risk for MDD in children of depressed parents has been estimated to

range from 15% (Orvaschel et al., 1988) to 45% (Hammen et al., 1990a). Factors in the depressed parent such as early onset and recurrence appear to confer the highest risk for MDD in children (e.g., Mufson et al., 1992; Orvaschel, 1990; Warner et al., 1995; Weissman et al., 1987, 1988). The risk for depression is also increased when both parents have mood disorders (e.g., Merikangas et al., 1988). Children of depressed parents are not only at high risk of developing depression but they are also at increased risk for general psychopathology, including anxiety and disruptive disorders (Biederman et al., 1991, 1995a; Hammen et al., 1990a; Keller et al., 1988; Mufson et al., 1992; Orvaschel et al., 1988; Warner et al., 1995; Weissman et al., 1987, 1988, 1992). These studies, as well as twin studies (e.g., Kendler et al., 1992), have suggested that what may be inherited is a vulnerability to depression and anxiety and that certain environmental stressors may be required for the manifestation of one of these disorders (Brown and Harris, 1993; Kendler, 1995; Plomin, 1994; Warner et al., 1995).

"Bottom-Up" Family Studies of Depressed Children. Age-unadjusted lifetime prevalence rates of depression in the first-degree relatives of depressed children and adolescents have been estimated to range from 20% to 46% (Kutcher and Marton, 1991; Livingston et al., 1985; Mitchell et al., 1989; Puig-Antich et al., 1989b; Strober, 1984; Todd et al., 1993; Williamson et al., 1995c). Family studies of adult-onset MDD have also consistently reported a two- to threefold increase in the lifetime rates of depressive disorders in the relatives of depressed subjects compared with normal controls (e.g., Gershon et al., 1982; Tsuang et al., 1985; Weissman et al., 1982, 1984a,b). However, there is an inverse relationship between age at onset and the density of familial aggregation of depression. Late-onset depression (≥ 60 years) is associated with the least risk for depression, and early-onset (≤ 20 years) is associated with the greatest risk for depression in family members (e.g., Puig-Antich et al., 1989b; Weissman et al., 1984b, 1988).

Dysthymic Disorder. A recent study of young adults with early-onset DD showed that DD and MDD patients had higher rates of MDD in their relatives compared with normal controls (Klein et al., 1995). The rate of DD, however, was significantly higher in the relatives of DD probands compared with MDD

and normal controls, suggesting that DD and MDD may have distinct patterns of familial transmission. Similar to the familial aggregation of MDD, patients with early-onset DD had higher rates of mood disorders in their relatives compared with patients with late-onset dysthymia (Klein et al., 1988a,b, 1995). Environmental factors seem to play a role. For example, chaotic family environments have been associated with an increased risk for dysthymia in offspring of parents who have MDD (Warner et al., 1995).

Family Environment

Studies of depressed adults (recalling their early family relationships), offspring of depressed parents, and depressed youths have shown that their family interactions are characterized by more conflict, more rejection, more problems with communication, less expression of affect, less support, and more abuse than are the family interactions of normal controls (e.g., Kaufman, 1991; McCauley and Myers, 1992). Nevertheless, the few studies that have included psychiatric control groups have questioned the specificity of these findings (Downey and Coyne, 1990; Goodman and Brumley, 1990; Stubbe et al., 1993), and it is possible that family conflict may reflect a coping strategy in which parents attempt to control the child's disruptive behaviors (Asarnow et al., 1994). Furthermore, it appears that there is an increased risk for any psychiatric disorder in children of mothers who had poor baseline functioning, independent of the mother's psychiatric status (Lee and Gotlib, 1991; Mufson et al., 1994).

The mechanisms by which abnormal family interactions might increase the risk of developing depression are not yet elucidated. One possible mechanism is that the abnormal early interactions between mother and child may cause children to develop patterns of handling stress that predispose them to depression. Parents may be teaching their children to "give up" when facing a stressful task, and not modeling adaptive ways to regulate negative affect (Garber and Hilsman, 1992). Family conflicts may also be an unspecific stress that triggers depression in individuals susceptible to it. Other factors such as lack of affect or irritability expressed toward the child, family conflicts, and abuse may also contribute to the child's increased vulnerability to depression or other psychopathology (e.g., Adrian and Hammen, 1993; Billing and Moos, 1986). Finally, it is important to indicate that depressed children

may also generate conflicts that will contribute to the maintenance of their parents' and their own depression (Hammen, 1991; Hammen et al., 1990b) or create conflicts in an otherwise normally functioning family.

Stressful Life Events

Several studies of depressed adults have also consistently shown, using different methodologies, that 60% to 70% have experienced one or more "severe" stressful life event(s) (particularly losses) in the year prior to the onset of the MDD (e.g., Brown and Harris, 1989, 1993; Frank et al., 1994). Several cross-sectional studies using both clinical and community samples of depressed children and adolescents have also found a modest but significant relationship between stressful life events and depression (e.g., Garber and Hilsman, 1992; Williamson et al., 1995a).

Specific events, including loss, divorce, bereavement, exposure to suicide alone, or together with other risk factors (e.g., lack of support), have been associated with the onset of depression (Brent et al., 1993c,d; Reinherz et al., 1993; Weller et al., 1991). For example, women who lost a parent before the age of 17 and had poor parental care were found to be at risk for having poor self-esteem, getting married early, and having children at an earlier age. These women were found to be at increased risk of developing depression when exposed to stressful life events, but the depression could be prevented by having a supportive spouse (Brown and Harris, 1993). Exposure to suicide is a severe, stressful event which has been associated with a threefold increase in acute and recurrent MDD in friends, siblings, and mothers of the suicide victims (Brent et al., 1988, 1993c,d). The risk of developing depression appears to be proportional to the closeness to the victim and the intensity of the exposure. Also, factors such as history of additional interpersonal losses, additional stressors, family psychiatric history, and prior psychopathology, including depression, increase the risk for depression (Brent et al., 1988, 1993c,d).

A major limitation for stressful life event research in early-onset depression is that much of the research has focused on cross-sectional correlational data obtained from self-report checklists, making it difficult to establish a causal relationship (events could be either a cause or a result of the depression). Studies using adaptations of the adult Life Event and Difficulty

Schedule Interview (Brown and Harris, 1989) for children and adolescents have reported significantly more severe and nonsevere stressful life events (especially in areas of romantic relationships, education, relationships with friends or parents, work, and health) in depressed youths 12 months prior to the onset of their depression compared with normal controls (Birmaher et al., 1995; Goodyer et al., 1985, 1988).

Negative Cognitive Style

Depression has been associated with low self-esteem, high self-criticism, significant cognitive distortions, and a feeling of lack of control over negative events (e.g., Beck, 1987; Garber and Hilsman, 1992). The "cognitive-diathesis" model proposes that individuals who are exposed to stressful events and who have negative styles of interpreting and coping with stress are at high risk of developing depressive symptoms (e.g., Garber and Hilsman, 1992). Although the mechanisms by which children and adolescents develop negative cognitive styles are not yet established, studies have suggested that certain factors, such as modeling significant others, perfectionistic standards, criticism, rejection, and experiences with uncontrollable stressful life events, may play a role (Garber and Hilsman, 1992).

Depressed children and adolescents, drawn from community and clinical samples, have been shown to have increased cognitive distortions, negative attributions, hopelessness, tendency to attribute outcomes to external noncontrollable causes, social skill deficits, and low self-esteem compared with nonaffective psychiatric and normal controls (e.g., Asarnow and Bates, 1988; Garber and Hilsman, 1992; Garber et al., 1993; Gladstone and Kaslow, 1995; Gotlib et al., 1993; Hammen, 1990; Marton et al., 1993b; Marton and Kutcher, 1995). Some studies have suggested that negative cognitive style seems to be specific for depression, in particular for more severe depressions (Marton et al., 1993a; Marton and Kutcher, 1995), but others not (e.g., Kazdin et al., 1983, 1986). Also, it is unclear whether the depressogenic cognitive style is a trait or a state characteristic (Asarnow and Bates, 1988; Hammen et al., 1986a,b). Longitudinal investigations have shown that after remission, depressed children have lower self-esteem, which in turn predicts future episodes of depression (e.g., Gotlib et al., 1993; Hammen, 1988; Marton et al., 1993a; Nolen-Hoeksema et al., 1992). Since lower self-esteem could be a lingering "scar" of

the depressive episode, investigations including patients at high risk for depression will help shed further light on this important area.

Recently, longitudinal studies of nonreferred school children have reported that a negative cognitive style predisposes children to experience a prolonged dysphoric mood when the child is exposed to stressors such as receiving poor report cards or being rejected by peers (Garber and Hilsman, 1992; Hilsman and Garber, 1995; Nolen-Hoeksema et al., 1992). However, this negative cognitive style seems to emerge mainly during early adolescence (Nolen-Hoeksema et al., 1992; Turner and Cole, 1994), or it may be the product of repeated subsyndromal or clinical episodes of depression. The above findings provide further evidence of the validity of the "cognitive-stress" model of depression. Nevertheless, studies examining the cognitive-stress diathesis in clinical samples are needed.

BIOLOGICAL MARKERS

Growth Hormone Studies. Like adults, depressed children have been found to hyposecrete growth hormone (GH) after various pharmacological challenges including insulin-induced hypoglycemia, oral clonidine, L-dopa, desmethylinipramine, and growth hormone-releasing hormone (e.g., Jensen and Garfinkel, 1990; Ryan et al., 1994). Furthermore, blunted GH response to provocative insulin-induced hypoglycemia has been reported to persist upon MDD remission, suggesting that this finding may be a "trait" or "scar" marker for MDD (Ryan et al., 1994). The dysregulation in GH secretion in depression may reflect changes in the central noradrenergic receptors, but also it may be secondary to changes in other neurotransmitters, somatomedins, and somatostatin, which have been reported to be altered in some depressed patients (Ryan et al., 1994). It is not clear yet whether the above findings are specific for depression or are associated with psychopathology in general.

Findings as to the control of nocturnal secretion of GH without stimulation are conflicting. Earlier studies of children (e.g., Puig-Antich et al., 1984) and adolescents (Kutcher et al., 1988, 1991) suggested that there may be a relative hypersecretion during sleep, while a recent study (De Bellis et al., 1996) has failed to replicate this finding. Another study of adolescents

found no differences in GH secretion between depressed and normal controls during the night, but a subgroup of suicidal, depressed adolescent inpatients showed decreased nocturnal GH secretion (Dahl et al., 1992b). A reexamination of nocturnal GH secretion in depressed children (e.g., Puig-Antich et al., 1984) suggested that stressful life events may contribute to elevated nocturnal GH secretion in this population (Williamson et al., in press).

Serotonergic Studies. Several biological investigations in adults with MDD have suggested that dysregulation of the central serotonergic function may be a vulnerability factor for the development of depression (Maes and Meltzer, 1995). To date, only one child study has shown that children with early-onset MDD have significantly lower cortisol levels than normal children after infusion of L-5-hydroxytryptophan (Ryan et al., 1992a). In addition, depressed female children appear to secrete significantly more prolactin compared with normal females and depressed and normal males (Ryan et al., 1992a).

Hypothalamic-Pituitary-Adrenal Axis. Evidence of hypothalamic-pituitary-adrenal axis dysregulation has been observed infrequently and inconsistently in studies of depressed children and adolescents, compared with studies of depressed adults. Investigations of baseline plasma cortisol secretion (24-hour or nocturnal sampling) have not found significant differences between depressed outpatients and normal control children and adolescents (Birmaher et al., 1992a,b; Dahl et al., 1991b; Kutcher et al., 1991; Puig-Antich et al., 1989a).

There have been numerous studies of the dexamethasone suppression test in depressed children (10 studies in inpatient settings and 4 outpatient studies) and depressed adolescents (11 inpatient samples and 2 outpatient studies) (Casat et al., 1989; Dahl et al., 1992a). The results, summarized across studies, indicate the following: (1) the sensitivity of the dexamethasone suppression test was higher in inpatients than in outpatients (61% versus 29% sensitivity); (2) the sensitivity was somewhat better in depressed children than in depressed adolescents (58% versus 44%); and (3) the specificity compared to other psychiatric controls for child inpatients was approximately 60%, whereas for adolescent inpatients it was approximately 85%. A number of factors have been discussed in relation to the variance of findings across studies including suicidality, differences in dexamethasone metabolism,

and variations in the stress of procedures (such as venipunctures for blood samples) and hospitalization stress (Birmaher et al., 1992a,b; Dahl et al., 1992a).

Studies in adults with MDD have consistently shown blunted adrenocorticotrophic hormone (ACTH) with normal cortisol plasma levels after the administration of corticotropin releasing hormone (CRH), particularly in inpatient and melancholic samples (for a review see Birmaher et al., 1996). A recent study found no significant differences between prepubertal children with MDD and normal controls in baseline or post-CRH stimulation values of either cortisol or ACTH (Birmaher et al., 1996). However, the depressed inpatients and the melancholic subgroups were found to secrete significantly less overall ACTH. Abnormalities in ACTH secretion in response to CRH have also been observed in abused children, with the nature of the ACTH disturbances affected by both past history of abuse and current stressors (Kaufman et al., 1993, 1995; De Bellis et al., 1994).

Sleep Studies. Despite frequent subjective complaints about disturbed sleep among depressed children and adolescents (Ryan et al., 1987), objective EEG studies have not found consistent sleep changes paralleling adult MDD studies. Among the four studies of children (Dahl et al., 1991a; Emslie et al., 1990; Puig-Antich et al., 1982; Young et al., 1982), only one study found decreased rapid eye movement (REM) latency and increased sleep latency in an inpatient sample of depressed children (Emslie et al., 1990). Among studies of depressed adolescents (Appleboon-Fondu et al., 1988; Dahl et al., 1990, 1996; Emslie et al., 1994; Goetz et al., 1987; Kahn and Todd, 1990; Kutcher et al., 1992; Lahmeyer et al., 1983), five reported prolonged sleep latency, four reduced REM latency, and three decreased sleep efficiency in the MDD subjects. To date, no studies have found any differences in delta sleep.

Greater rates of sleep changes have been observed in inpatient adolescent samples and also in association with psychosis, suicidality, and endogenous MDD subtypes (Dahl et al., 1990; Emslie et al., 1994; Kutcher et al., 1992; Naylor et al., 1990). To date, only one study has assessed the sleep in children after recovery from depression (Puig-Antich et al., 1983). This study showed improvement in sleep efficiency but reduced REM latency. A recent study showed that depressed

adolescents without any stressful life events had significantly lower REM latencies compared with normal controls without stressful life events, suggesting the need to incorporate measures of environmental stress in EEG sleep studies (Williamson et al., 1995b).

CONCLUSIONS

Several lines of evidence provide support for the validity of childhood MDD and DD and their continuity with the adult depressive disorders. Early-onset MDD and DD are frequent, familial, and recurrent disorders that are accompanied by other psychiatric disorders, in particular anxiety and disruptive disorders. Both disorders increase the risk for substance abuse, suicidal behavior, and poor psychosocial and functional outcome. MDD also increases the risk for bipolar disorder and DD the risk for the development of future MDD episodes. There is a secular increase in the prevalence of MDD, and it appears that it is occurring at an earlier age in successive cohorts, underscoring the necessity for the early identification and treatment of these disorders.

Much progress has been made toward identifying correlates of onset, course, and recurrence of depression in children and adolescents (for further review see Brent et al., 1995). Further understanding of the biological underpinnings and the interrelationships among psychosocial and psychobiological factors in early-onset depression is needed (e.g., Birmaher et al., 1994; De Bellis et al., 1994; Kaufman et al., 1995; Williamson et al., 1995a,b, in press). Risk factors such as parental psychopathology, family conflict, peer support, and poor coping skills appear to increase the risk for developing mood disorders (Billing and Moos, 1986; Mufson et al., 1994; Rutter, 1990). It is important to mention that these factors seem to be nonspecific correlates of child psychopathology (Biederman et al., 1995b; Blantz et al., 1991; Rutter, 1990); however, as Garber and Hollon (1991) suggested, "it is possible for a particular variable to be nonspecific yet still causal if the variable is part of a larger multivariable causal process" (p. 129). Furthermore, factors that are associated with the onset of more than one type of psychopathology may be more important to target since they may reduce the risk for the development of disturbances in multiple areas. Finally, it is important

to emphasize that depression is a heterogeneous disorder, and it is likely that various processes may operate in its pathogenesis.

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